EET202

HINES & TRANSFORMERS

MODULE:2

e:2 – Losses and Power Flow

e performance characteristics of DC generators

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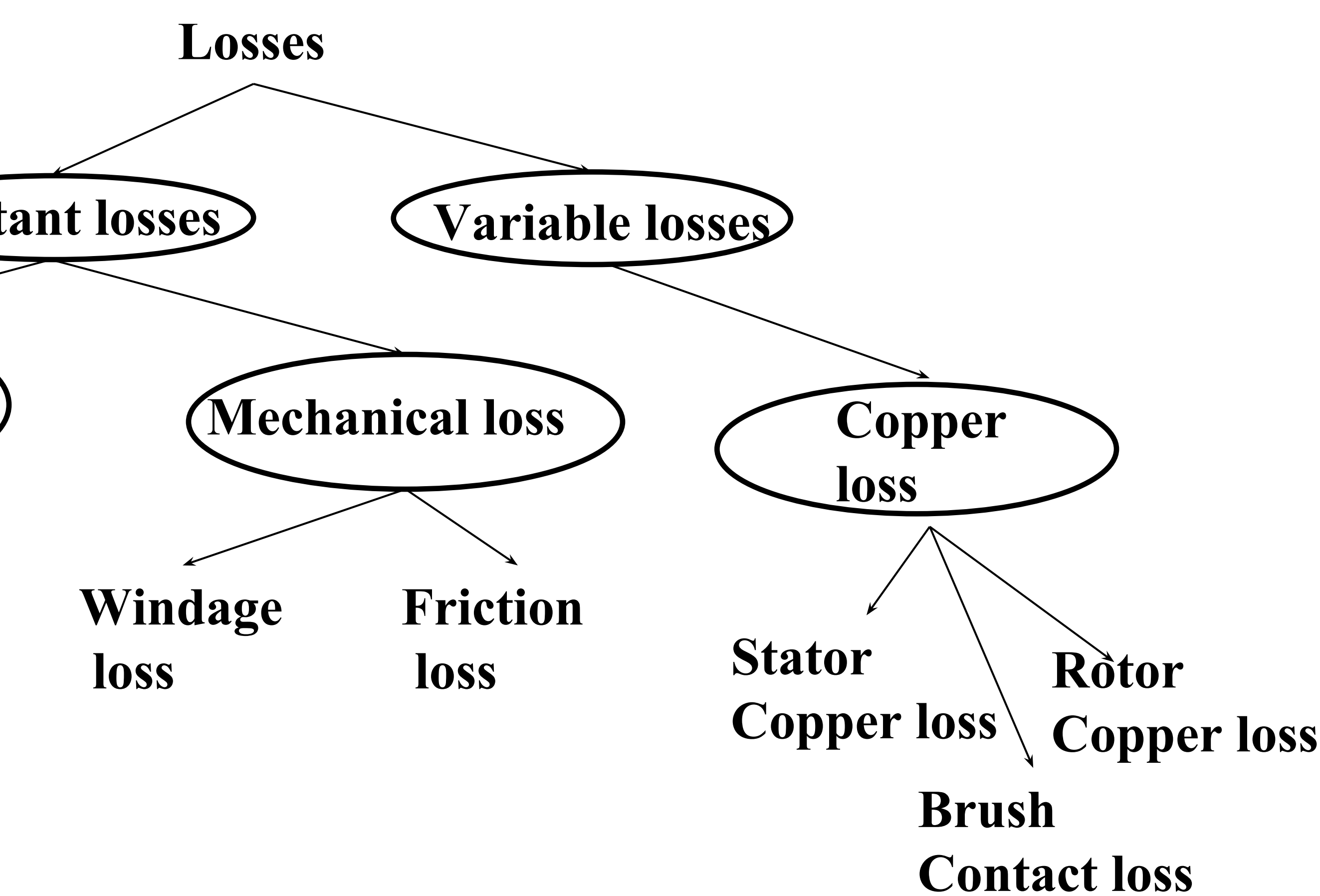
ses in DC Generator



ses in DC Generator

<u>sses</u>

enerator which remain constant for all
l constant losses.

<u>ses</u>

enerator which vary with load are called

ses in DC Generator

ses

in the armature core of a DC machine and are
of armature in the magnetic field of the poles.

t Loss

oss

oss

eded to overcome mechanical friction and wind
aft.

s

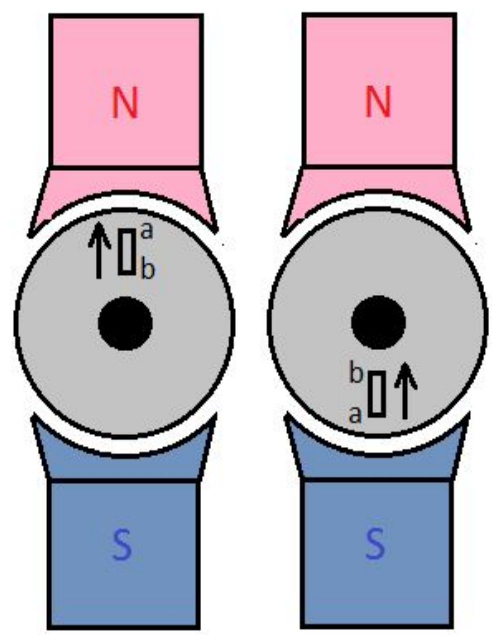
ses in DC Generator

the armature of dc machine since
e armature is subjected to magnetic
asses under successive poles.

ece 'ab' of armature as shown in

is under N-pole, the magnetic lines
After half a revolution, the same
-pole and magnetic lines pass from
tism in the iron piece is reversed.

ontinuously the molecular magnets
e, some amount of power has to be
l Hysteresis loss.



$$W_h = \eta B_m^{1.6} f V$$

y in the armature, f *: Frequency of flux reversals*

n m^3, η *: Steinmetz hysteresis co-efficient*

ses in DC Generator

core is a conducting material, an emf is core according to Faraday's law of uction, in addition to the voltage induced in ctors.

uce circulating current in the core.

ent is known as eddy current.

esponsible for doing any work.

to flow of this current is called eddy current

pears as heat which raises the temperature owers its efficiency.

Where,

K_e: Constant, B_m: Maximum flux density (Wb/m^2),
f: Frequency of magnetic flux reversals,
t: Thickness of lamination, V: Volume of core

ses in DC Generator

ses

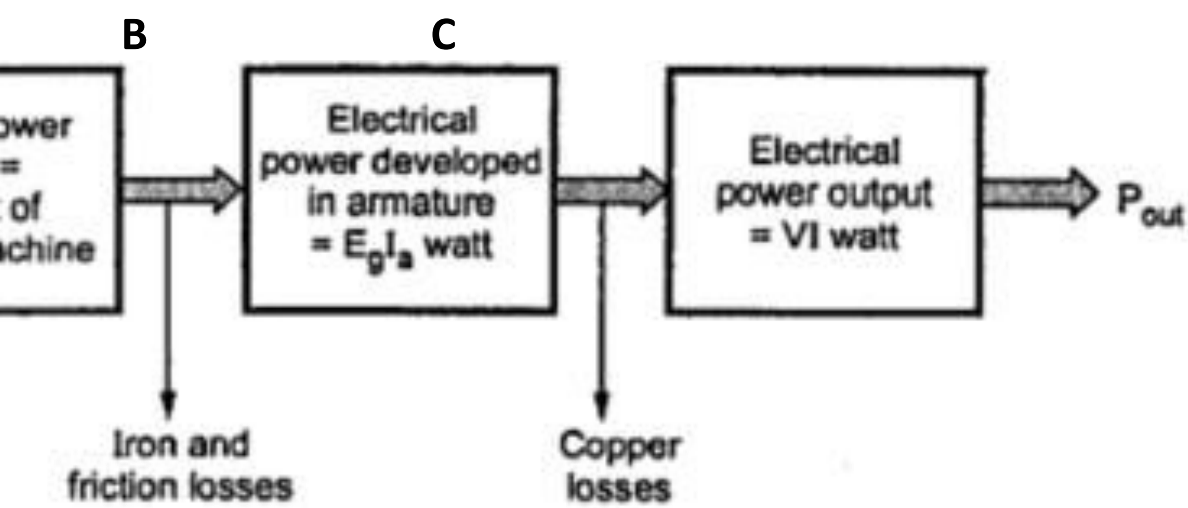
cur due to voltage drop in windings when

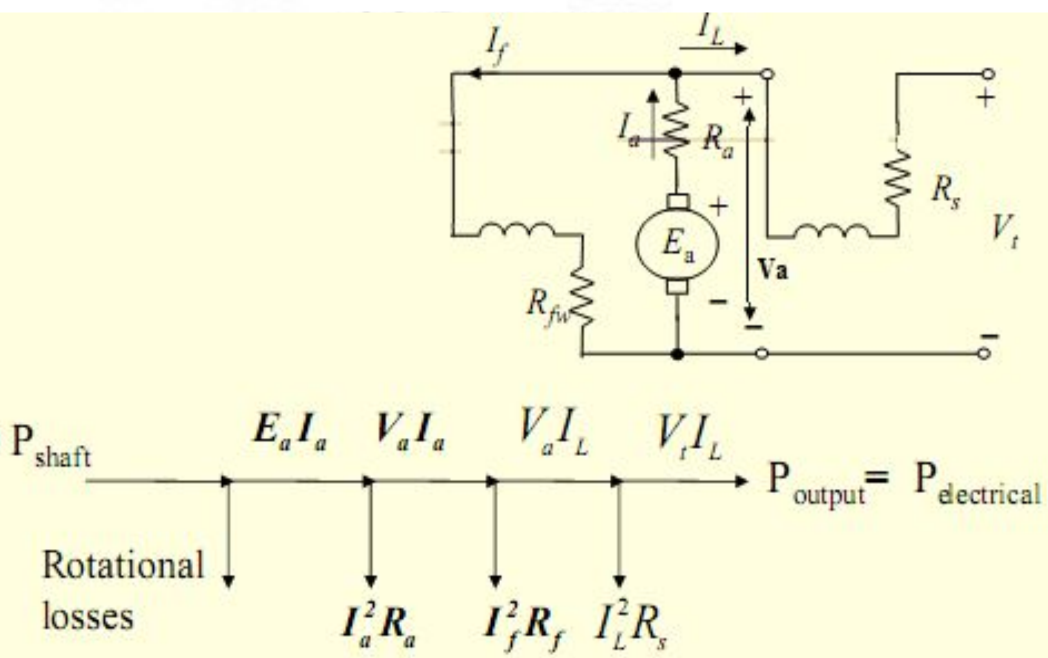
pper loss = $I_a^2 R_a$

opper loss = $I_{sh}^2 R_{sh}$

opper loss = $I_{se}^2 R_{se}$

wer Flow Diagram





Efficiencies

efficiency, η_m = power generated in
echanical power input

ficiency, η_e = electrical power output/
rated in armature

iency, η = $\eta_m \eta_e$ = electrical power
chanical power input

n for Maximum Efficiency

DC generator is not constant but varies with hunt generator delivering a load current I_L at a /.

t = VI_L

= output + losses

+ variable losses + constant losses VI_L

$VI_L + I_a^2 R_a + W_c$

$VI_L + (I_L + I_{sh})^2 R_a + W_c$

current I_{sh} is generally small as compared to I_L be neglected.

= $VI_L + I_L{}^2 R_a + W_c$

n for Maximum Efficiency

ut/input = $\dfrac{VI_L}{VI_L + I_L^2 R_a + W_c}$

$$\frac{1}{1 + \frac{I_L R_a}{V} + \frac{W_c}{VI_L}}$$

aximum when denominator is minimum.

0

$$\frac{R_a}{V} = \frac{W_c}{VI_L^2} \Rightarrow I_L^2 R_a = W_c$$

Constant loss (since $I_L \approx I_a$)

orresponding to maximum efficiency is given by:

$$I_L = \sqrt{\frac{W_c}{R_a}}$$

y & Losses vs. Load Current